Collective three-body interactions enable a robust speedup in quantum sensing: Supplemental Materials

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S1. MULTI-BODY INTERACTIONS IN CAVITY-QED SYSTEM

In this section, we sketch the derivation of multi-body interactions in cavity-QED-based matter-wave interferometers [S1–S5], with full details provided in the supplementary material of Ref. [S6].

As illustrated in Fig. 1 of the main text, a single ground-state level $|g\rangle$ within the atomic manifold is coupled to an excited state $|e\rangle$, separated by energy ω_a, via a single cavity mode $\hat{a}$ of frequency ω_c with finite decay rate κ. Two classical dressing lasers are applied to the cavity with amplitudes $\epsilon_{1,2}$ and frequencies $\omega_{p1,2}$. The cavity detuning from the e to g atomic transition is $\Delta_a = \omega_c - \omega_a$. The atom-cavity coupling is characterized by a peak single-photon Rabi frequency $2g$.

We start with the full atom-cavity Hamiltonian, under the rotating frame of the cavity field $\hat{H}_0 = \hbar\omega_c \hat{a}^\dagger \hat{a} + \hbar\omega_c \int dz\ \hat{\psi}_e^\dagger(z)\hat{\psi}_e(z)$, given as $\hat{H} = \hat{H}_{\text{atom}} + \hat{H}_{\text{light}} + \hat{H}_{\text{int}}$ with the following second quantized form:

$$\hat{H}_{\text{light}} = \hbar\left(\epsilon_1 \hat{a}^\dagger e^{-i(\omega_{p1}-\omega_c)t} + \epsilon_2 \hat{a}^\dagger e^{-i(\omega_{p2}-\omega_c)t} + \text{h.c.}\right) \tag{S1}$$

$$\hat{H}_{\text{atom}} = \sum_{\tau=g,e} \int \hat{\psi}_\tau^\dagger(Z)\frac{\hat{p}^2}{2m}\hat{\psi}_\tau(Z)\, dZ - \hbar\Delta_a \int \hat{\psi}_e^\dagger(Z)\hat{\psi}_e(Z)\, dZ \tag{S2}$$

$$\hat{H}_{\text{int}} = \hbar \int 2g\cos kZ \left[\hat{a}\hat{\psi}_e^\dagger(Z)\hat{\psi}_g(Z) + \hat{a}^\dagger \hat{\psi}_g^\dagger(Z)\hat{\psi}_e(Z)\right] dZ. \tag{S3}$$

Here, $\hat{\psi}^\dagger_{e(g)}(Z)$ are the field creation operator for the atom at position Z in $|e\rangle$ ($|g\rangle$) state. If $\Delta_a \gg g\sqrt{N\langle \hat{a}^\dagger \hat{a}\rangle}$ and $\Delta_a \gg \gamma$, where γ denotes the spontaneous emission rate of the excited state $|e\rangle$, we can adiabatically eliminate the excited state and obtain the dispersive atom-cavity coupling Hamiltonian acting only on the ground levels:

$$\hat{H}_S = \int \hat{\psi}_g^\dagger(Z)[\frac{\hat{p}^2}{2m} + \frac{4\hbar g^2}{\Delta_a}\cos^2(kZ)\hat{a}^\dagger\hat{a}]\hat{\psi}_g(Z)\, dZ + \hbar\left(\epsilon_1 \hat{a}^\dagger e^{-i(\omega_{p1}-\omega_c)t} + \epsilon_2 \hat{a}^\dagger e^{-i(\omega_{p2}-\omega_c)t} + \text{h.c.}\right). \tag{S4}$$

We then perform a Fourier transformation on the atomic field operator $\hat{\psi}_g(Z)$ to the momentum space as $\hat{\psi}_g(Z) = \int_{-\infty}^{\infty} \hat{\psi}(p) e^{ipZ/\hbar}\, dp$, here $\hat{\psi}(p)$ annihilates an atom with momentum p. Then Hamiltonian can be rewritten as:

$$\hat{H}_S = \int [\frac{p^2}{2m}\hat{\psi}^\dagger(p)\hat{\psi}(p) + \frac{\hbar g^2}{\Delta_a}\hat{a}^\dagger\hat{a}(\hat{\psi}^\dagger(p+2\hbar k)\hat{\psi}(p) + \hat{\psi}^\dagger(p-2\hbar k)\hat{\psi}(p))]\, dp + \hbar\left(\epsilon_1 \hat{a}^\dagger e^{-i(\omega_{p1}-\omega_c)t} + \epsilon_2 \hat{a}^\dagger e^{-i(\omega_{p2}-\omega_c)t} + \text{h.c.}\right). \tag{S5}$$

Notably, the spatially inhomogeneous atom-cavity coupling becomes a homogeneous coupling between momentum states separated by $\pm 2\hbar k$.

In the experiment, the initial state consists of a wave packet with average momentum $p_0 - \hbar k$ and width σ_p, prepared from an array of thermal atoms via velocity selection. We then drive a superposition between $|\Downarrow_p\rangle \equiv |p_0 - \hbar k + p\rangle$ and $|\Uparrow_p\rangle \equiv |p_0 + \hbar k + p\rangle$, where k is the wave vector of the cavity mode and $\sigma_p \ll 2\hbar k$. We define the pseudospin-1/2 operators in terms of annihilation operators $\hat{\psi}_p^\Downarrow = \hat{\psi}(p_0 - \hbar k + p)$ for the $|\Downarrow_p\rangle$ momentum state and $\hat{\psi}_p^\Uparrow = \hat{\psi}(p_0 + \hbar k + p)$ for the $|\Uparrow_p\rangle$ momentum state:

$$\begin{aligned}
\hat{s}_{x,p} &= \frac{1}{2}\left((\hat{\psi}_p^\Uparrow)^\dagger \hat{\psi}_p^\Downarrow + (\hat{\psi}_p^\Downarrow)^\dagger \hat{\psi}_p^\Uparrow\right), \\
\hat{s}_{y,p} &= \frac{1}{2i}\left((\hat{\psi}_p^\Uparrow)^\dagger \hat{\psi}_p^\Downarrow - (\hat{\psi}_p^\Downarrow)^\dagger \hat{\psi}_p^\Uparrow\right), \\
\hat{s}_{z,p} &= \frac{1}{2}\left((\hat{\psi}_\Uparrow^p)^\dagger \hat{\psi}_\Uparrow^p - (\hat{\psi}_\Downarrow^p)^\dagger \hat{\psi}_\Downarrow^p\right),
\end{aligned} \tag{S6}$$

In terms of them we define collective spin operators

$$\hat{S}_\alpha = \sum_n \hat{s}_{\alpha,p} \quad \alpha \in [X, Y, Z, +, -]. \tag{S7}$$

Then the Hamiltonian in Eq. (S5) can be written in the pseudospin-1/2 operator,

$$\hat{H}_S/\hbar = \omega_z \hat{S}_z + g_{\text{eff}} \hat{a}^\dagger \hat{a} \left(\hat{S}_+ + \hat{S}_-\right) + \left(\epsilon_1 e^{-i(\omega_{p1}-\omega_c)t} + \epsilon_2 e^{-i(\omega_{p2}-\omega_c)t}\right) \hat{a}^\dagger + \text{h.c.}, \tag{S8}$$

where $\omega_z = 2(p_0 - \hbar k)k/M$ represents the kinetic energy difference between the central momentum states $|p_0 \pm \hbar k\rangle$, and $g_{\text{eff}} = g^2/\Delta_a$. We ignore the single-particle inhomogeneity term $\sum_p \omega_p \hat{s}_{z,p}$ [S2, S6], arising from the finite momentum spread as the collective interaction timescale is much faster than the Doppler dephasing. In the experiment, Bragg lasers are applied to prepare atoms in a superposition of two momentum states centered around $p_0 \pm \hbar k$, which we encode as the pseudo-spin states $|\uparrow\rangle \equiv |p_0 + \hbar k\rangle$ and $|\downarrow\rangle \equiv |p_0 - \hbar k\rangle$, with an energy splitting ω_z.

We set $\omega_{p2} - \omega_{p1} = 3\omega_z$ in the experiment to satisfy the resonance conditions for three-body dynamics between $|p_0 \pm \hbar k\rangle$ momentum state [S2], and when the detunings $\Delta_{c2} = \omega_{p2} - \omega_z - \omega_c$ and $\Delta_{c1} = \omega_{p1} + \omega_z - \omega_c$ satisfy $|\Delta_{c1,2}| \gg g_{\text{eff}} |\alpha_{1,2}| \sqrt{N}$, with $\alpha_{1,2} = \frac{\epsilon_{1,2}}{i\kappa/2 + (\omega_{p1,p2} - \omega_c)}$ denoting the intracavity classical field amplitudes, we can adiabatically eliminate the cavity-field fluctuations and obtain an effective three-body interactions (3BIs):

$$\hat{H}_3 = \chi_3 \hat{S}_+^3 + \chi_3^* \hat{S}_-^3, \quad \chi_3 \approx \frac{g_{\text{eff}}^3}{\Delta_{c1}\Delta_{c2}} \alpha_1^* \alpha_2, \tag{S9}$$

The phase of χ_3 can be tuned by adjusting the relative phase between the two dressing lasers. In the following discussion, we consider a symmetric pumping configuration by setting $\alpha_1 = \alpha_2 \equiv \alpha$ and $-\Delta_{c1} = \Delta_{c2} \equiv \Delta_c$, and take χ_3 to be real.

The effective Hamiltonian is also accompanied by two collective superradiant decay channels [S1], which arise from a process in which an atom absorbs a pump photon and subsequently emits a cavity photon that leaks out of the cavity at a finite rate κ:

$$L_+ = \sqrt{\Gamma^+} \hat{S}_+, \quad L_- = \sqrt{\Gamma^-} \hat{S}_-, \quad \Gamma^\pm = g_{\text{eff}}^2 |\alpha|^2 \frac{\kappa}{\Delta_c^2}. \tag{S10}$$

With the two superradiant decay rates are balanced under above symmetric pumping configuration [S2, S4]. Single-particle dissipation is modeled as a spin-flip process with the jump operators [S3]

$$\hat{L}_{s,i} = \sqrt{\gamma_e} \hat{s}_{-,i} \quad \gamma_e \approx 2 g_{\text{eff}} |\alpha|^2 \frac{\gamma}{\Delta_a}. \tag{S11}$$

In Sec. S3 B, we also benchmark with a more realistic single-particle atom loss processes, which account for light induced scattering into momentum states outside the the pseudo-spin states participating in the matter-wave interferometer.

S2. UNITARY DYNAMICS

A. Semi-classical analysis of three-body interactions

In this part, we estimate the time scales for the dynamics via the semi-classical phase space method, introduced in Fig. [S7]. We start with the Heisenberg equations of motion for collective spin operators, $d\langle \hat{S}_\alpha \rangle = i\langle [\hat{H}_3, \hat{S}_\alpha] \rangle$ with $\alpha = x, y, z$. As discussed in the main text, we define the classical variables $(x, y, z) \equiv (\langle \hat{S}_x \rangle, \langle \hat{S}_y \rangle, \langle \hat{S}_z \rangle)/(N/2)$, with the mean-field equations of motion:

$$\frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{3}{2} \chi_3 N^2 \begin{pmatrix} -2xyz \\ z(y^2 - x^2) \\ 3x^2 y - y^3 \end{pmatrix}. \tag{S12}$$

There exist six stable fixed points on the equators $(\cos\varphi, \sin\varphi, 0)$ with $\varphi = n\pi/3, n \in \mathbb{Z}$, and two unstable fixed points on the north and south pole $(0, 0, \pm 1)$ [S2]. These structure are consistent with the symmetry of the Hamiltonian: a

π rotation about the $\hat{y}$ axis, $e^{i\pi\hat{S}_y}\hat{H}_3 e^{-i\pi\hat{S}_y} = -\hat{H}_3$, or a $\pi/3$ rotation about the $\hat{z}$ axis, $e^{i\pi/3\hat{S}_z}\hat{H}_3 e^{-i\pi/3\hat{S}_z} = -\hat{H}_3$, effectively flips the sign of χ_3. In the discussion below, we focus on the case with $\chi_3 > 0$.

As discussed in the main text, the variance around the unstable point at the north pole, is evolving inwards along $\varphi = \pi/6 + 2n\pi/3$ and outwards along $\varphi = \pi/2 + 2n\pi/3$, with $n \in \mathbb{Z}$. Without loss of generality, we study the dynamics along the great circle with $x = 0$, thus $y = \sqrt{1 - z^2}$ to obtain:

$$\frac{dz}{dt} = -\frac{3}{2}\chi_3 N^2 (1 - z^2)^{3/2}. \tag{S13}$$

To study the time for the peak Quantum Fisher information (QFI) with initial coherent spin state prepared along $\hat{z}$ axis, we apply the method in Ref. [S7], where the point at the edge of the uncertainty patches $b_i = (0, \sqrt{1/N}, \sqrt{1 - 1/N})$, evolves towards the equators $b_f = (0, 1, 0)$. Therefore, we find:

$$\frac{3}{2}\chi_3 N^2 t_{\mathrm{opt},3} = -\int_{\sqrt{1-1/N}}^{0} \frac{dz}{(1 - z^2)^{3/2}} \Rightarrow t_{\mathrm{opt},3} = \frac{2}{3\chi_3 N^{3/2}}, \tag{S14}$$

assuming $N \gg 1$. The estimated optimal time matches the numerical result in Fig. 2 in the main text.

B. Generation of GHZ-like states

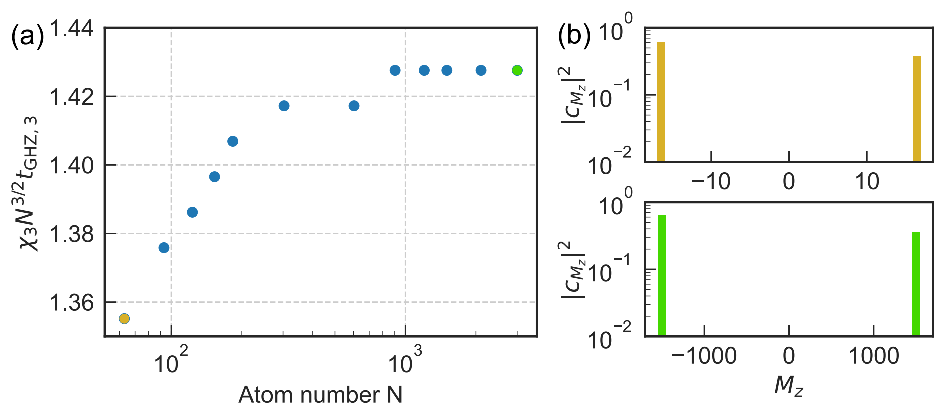

FIG. S1. (a) Optimal time $t_{\mathrm{GHZ},3}$ for generating a GHZ-like state as a function of atom number N. The scaled time $\chi_3 N^{3/2} t_{\mathrm{GHZ},3}$ saturates to 1.42 in the large-N limit. (b) Population distribution of the GHZ-like state at the optimal time for $N = 33$ (brown) and $N = 3003$ (green). The populations outside the $M_z = \pm N/2$ Dicke states remain below 10^{-2}.

In Fig. S1(a), we numerically simulate the optimal time for generating a GHZ-like state as a function of the atom number, ranging from $N = 33$ to $N = 3003$ with $N = 6n + 3$, $n \in \mathbb{Z}$. We use the population $|c_{N/2}|^2 + |c_{-N/2}|^2$ as a metric to evaluate the prepared state. We find that as the atom number increases, the optimal time $\chi_3 N^{3/2} t$ saturates to approximately 1.42.

In Fig. S1(b), we show the spin distribution function for $N = 33$ (brown) and $N = 3003$ (green). For both cases, we observe that $|c_{N/2}|^2 \approx 0.62$ for $N = 33$ and $|c_{N/2}|^2 \approx 0.64$ for $N = 3003$. Meanwhile, the remaining populations at Dicke states with $M_z \neq \pm N/2$ are below 10^{-2}, and this behavior is consistent across all atom numbers simulated. Note that the prepared state is not an exact GHZ state with equal weights on $|c_{\pm N/2}| \neq 1/\sqrt{2}$. This is a consequence of the intrinsic non-equilibrium dynamics of the 3BIs model rather than a numerical artifact; however, the state features high QFI $F_Q = 0.96N^2$ remains useful for quantum sensing [S8] and other quantum information tasks [S9].

In Fig. 2(c) of the main text, we plot the spin Wigner distribution of the generated state $\rho_t = e^{-i\hat{H}_3 t}|\psi_0\rangle\langle\psi_0| e^{i\hat{H}_3 t}$ defined as:

$$W(\theta, \varphi) = \sum_{S=0}^{N} \sum_{M_z=-S}^{S} \mathrm{Tr}(\rho_t T^{\dagger}_{S,M_z}) Y_{S,M_z}(\theta, \varphi), \tag{S15}$$

where T_{S,M_z} are irreducible tensor operators forming a basis for spin observables, and Y_{S,M_z} are standard spherical harmonics on the Bloch sphere. This representation allows us to visualize the quantum state in spin phase space, with negative regions indicating nonclassical features.

C. Comparison with two-body interaction

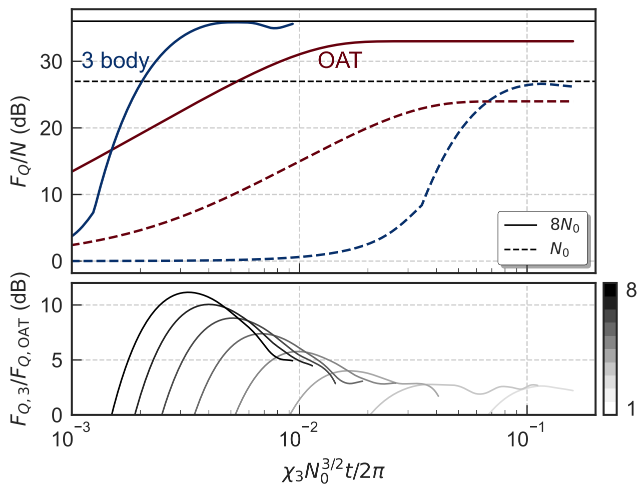

FIG. S2. Upper panel: Time evolution of the QFI under 3BIs (blue) and OAT (red). Dashed lines correspond to $N = 500$ atoms, while solid lines correspond to $N = 4000$ atoms. The vertical lines represent the corresponding Heisenberg limit $F_Q = N^2$. Lower panel: Ratio of QFI between 3BIs and OAT as a function of total atom number, ranging from $N = 500$ to $N = 4000$. The 3BIs yield over 10 dB peak enhancement in QFI compared to the OAT scheme at $N = 4000$. All simulations are performed with $\chi_2 = 10\chi_3/\sqrt{N}$.

In this section, we compare the performance of 3BIs with the well-known two-body one-axis twisting (OAT) model, or equivalently the all-to-all Ising interaction. As discussed in the main text, the two interaction strengths are related by $\chi_2 = \chi_3\sqrt{N}/\eta$. In Fig. S2, we fix $\eta = 0.1$ and examine the growth of the QFI for $N = 500$ (dashed lines) and $N = 4000$ (solid lines) in the upper panel. The blue curves correspond to the QFI under 3BIs, denoted $F_{Q,3}$, while the red curves show the QFI of the OAT model, $F_{Q,\mathrm{OAT}}$. The evolution time is rescaled using a reference atom number $N_0 = 500$. We find that the 3BIs reach the Heisenberg limit significantly faster than OAT. Moreover, as the atom number increases, $F_{Q,3}$ surpasses $F_{Q,\mathrm{OAT}}$ at progressively earlier times. In the lower panel, we plot the ratio $F_{Q,3}/F_{Q,\mathrm{OAT}}$ as a function of time for N ranging from 500 to 4000. For $N = 4000$, the QFI generated by 3BIs exceeds that of OAT by more than 10 dB. These results demonstrate that 3BIs enable substantially faster generation of metrologically useful quantum states compared with OAT dynamics.

D. Analytical solutions from truncated wigner approximations

In this section, we derive analytical results for the 3BIs dynamics using the semi-classical Truncated Wigner Approximation (TWA). For a pure state, the QFI corresponds to four times the variance along the phase accumulation direction, which in our case is $4\,\mathrm{Var}(\hat{S}_z)$. Fig. S3(a) shows the numerical results, where the TWA simulation (orange curve) and the analytical approximation from Eq. (S19) (green curve) both agree well with the exact solution (blue curve) during the initial growth of QFI. In the following, we detail the derivation of these results.

For the TWA simulation, we set at time $t = 0$, $z_0 = 1$, while x_0 and y_0 are sampled from a Gaussian distribution $W(x_0, y_0)$ with a mean of 0 and a variance of $1/N$. The mean-field equations of motion in Eq. (S12) conserve the spin length, The mean-field equations of motion in Eq. (S12) conserve the spin length $\langle\hat{S}_+\rangle\langle\hat{S}_-\rangle + \langle\hat{S}_z\rangle^2 = (N/2)^2 + S^2$, which leads to the classical relation $x_t^2 + y_t^2 + z_t^2 = 1 + s_0^2$. They also conserve the energy, $\langle\hat{S}_+\rangle^3 + \langle\hat{S}_-\rangle^3 = 2S^3\cos 3\varphi$ which gives $(x_t + iy_t)^3 + (x_t - iy_t)^3 = 2s_0^3\cos 3\varphi_0$. Here we define $s_0 = \sqrt{x_0^2 + y_0^2}$ and $\varphi_0 = \arctan(y_0/x_0)$ for the initial sampled (x_0, y_0). It turns out that s_0 follows a Rayleigh distribution $f(s_0) = s_0 N e^{-Ns_0^2/2}$ and φ_0 is uniformly distributed in $[0, 2\pi]$. With the above conservation laws, we find the first-order differential equation for z_t:

$$
\begin{aligned}
(\dot{z}_t)^2 &= (6\chi_3 N^2)^2[(1 + s_0^2 - z_t^2)^3 - s_0^6\cos^2 3\varphi_0] \\
\Rightarrow \mathrm{sign}(\sin 3\varphi_0)6\chi_3(N/2)^2 t &= \int_1^{z_t} \frac{dz_t}{\sqrt{(1 + s_0^2 - z_t^2)^3 - s_0^6\cos^2\varphi_0}}.
\end{aligned}
\tag{S16}
$$

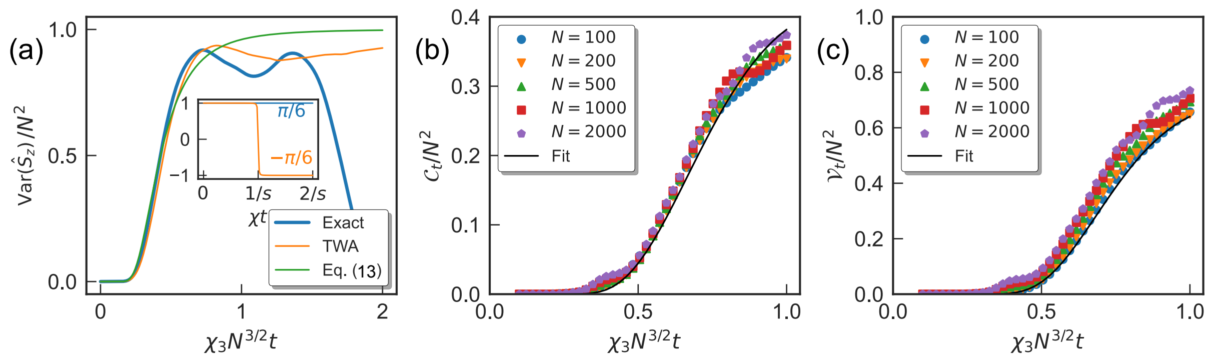

FIG. S3. (a) Growth of variance $\mathrm{Var}(\hat{S}_z)$ over time for $N = 500$ atoms, with the blue curve representing the exact solution, the orange curve representing the TWA result, and the green curve representing the approximated solution in Eq. (S19). The inset plots $z_t(s_0, \pm\pi/6)$ defined in Eq. (S17) with $s_0 = 0.01$. (b), (c) Rescaled $\mathcal{C}_t$ and $\mathcal{V}_t$ as functions of time for different atom numbers. The black curves represent the fitted results in Eq. (S26).

In general, an analytical solution for the above integral is complex for $\cos 3\varphi_0 \neq 0$. The numerical result is shown as the orange curve in Fig. S3(a). Instead, we can fix $\varphi_0 = \pm\pi/6$ to obtain:

$$\begin{aligned} z_t(s_0, \varphi_0 = \pi/6) &= \frac{1 + \tilde{\chi} t (s_0^3 + s_0)}{\sqrt{1 + 2\tilde{\chi} t s_0 + \tilde{\chi}^2 t^2 (s_0^4 + s_0^2)}} \approx 1 \\ z_t(s_0, \varphi_0 = -\pi/6) &= \frac{1 - \tilde{\chi} t (s_0^3 + s_0)}{\sqrt{1 - 2\tilde{\chi} t s_0 + \tilde{\chi}^2 t^2 (s_0^4 + s_0^2)}} \approx \begin{cases} 1 & \chi t < 1/s \\ -1 & \chi t > 1/s \end{cases} \end{aligned} \tag{S17}$$

with $\tilde{\chi} = 6\chi_3 (N/2)^2$, We plot the above two functions in the inset of Fig. S3(a). Then the TWA result can be approximated as follows:

$$\begin{aligned} \langle \hat{S}_z \rangle &\approx \frac{N}{2} \int_0^\infty ds_0 N s_0 e^{-N s_0^2/2} \frac{z_t(s_0, \pi/6) + z_t(s_0, -\pi/6)}{2} \\ &\approx \frac{N}{4} (1 + \int_0^{\frac{1}{t}} ds_0 N s_0 e^{-N s_0^2/2} - \int_{\frac{1}{t}}^\infty ds_0 N s_0 e^{-N s_0^2/2}) = \frac{N}{2} (1 - e^{-\frac{N}{2(\tilde{\chi} t)^2}}) \\ \langle \hat{S}_z^2 \rangle &\approx (\frac{N}{2})^2 \int_0^\infty ds_0 N s_0 e^{-N s_0^2/2} (\frac{z_t(s_0, \pi/6) + z_t(s_0, -\pi/6)}{2})^2 \approx (\frac{N}{2})^2, \end{aligned} \tag{S18}$$

with the variance is given by

$$\mathrm{Var}(\hat{S}_z) = (\frac{N}{2})^2 (2 e^{-\frac{2}{9(\chi_3 N^{3/2} t)^2}} - e^{-\frac{4}{9(\chi_3 N^{3/2} t)^2}}), \tag{S19}$$

and is plotted as the blue curve in Fig. S3(a).

E. Time-reversal protocol

In this section, we analyze the time-reversal protocol discussed in the main text. The system is initialized in the state $|\psi_0\rangle = |N/2, N/2\rangle$, followed by phase accumulation around the $\hat{S}_z$ axis by a small angle $\phi \ll 1$. The final measurement is performed by evaluating the observable $\hat{O}$. The expectation value of the observable after the entire sequence is given by:

$$\langle \hat{O} \rangle = \langle \psi_0 | \hat{V}^\dagger \hat{O} \hat{V} | \psi_0 \rangle \quad \hat{V} = e^{it\hat{H}_3} e^{-i\phi \hat{S}_z} e^{-it\hat{H}_3}. \tag{S20}$$

We can perform an expansion over the small angle ϕ to obtain:

$$\langle \hat{O} \rangle \approx \langle \psi_0 | \hat{O} | \psi_0 \rangle + \phi^2 (\langle \psi_0 | \hat{S}_z(t) \hat{O} \hat{S}_z(t) | \psi_0 \rangle - \frac{1}{2} \langle \psi_0 | \hat{S}_z^2(t) \hat{O} - \hat{O} \hat{S}_z^2(t) | \psi_0 \rangle), \tag{S21}$$

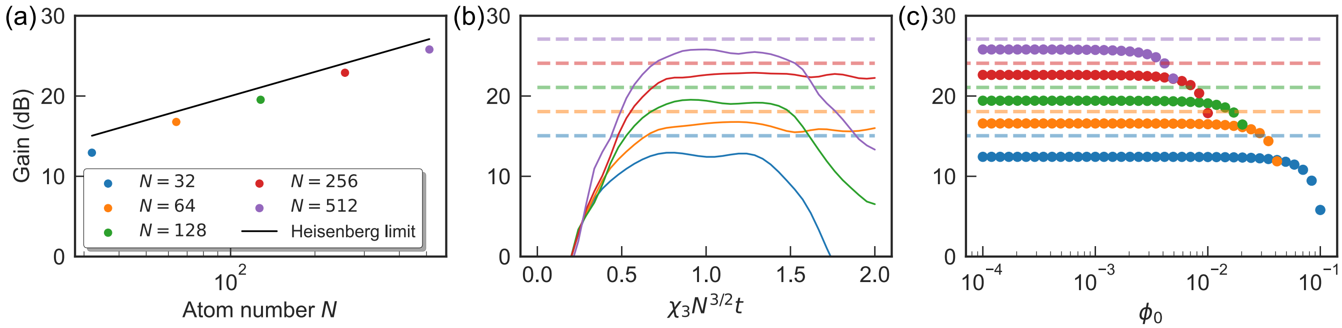

FIG. S4. Exact numerical simulation of the time-reversal protocol. (a) Metrological gain as a function of atom number, for $N = 32, \cdots, 512$ with fixed $\phi_0 = 10^{-4}$. The black line indicates the Heisenberg limit. (b) Metrological gain versus time for various N, with fixed $\phi_0 = 10^{-4}$. The dashed lines denote the Heisenberg limit for different atom numbers. (c) Dynamical range of the protocol, where ϕ_0 denotes the center phase around which the signal is measured.

where we define $\hat{S}_z(t) = e^{i\hat{H}_3 t}\hat{S}_z e^{-i\hat{H}_3 t}$.

For a final projection measurement $\hat{P} = |\psi_0\rangle\langle\psi_0|$ onto the initial state, which is an N-body operator, it is known that [S10, S11]

$$\langle\hat{P}\rangle \approx 1 - \phi^2 \mathrm{Var}(\hat{S}_z) = 1 - \frac{\phi^2 F_Q}{4}, \tag{S22}$$

where F_Q denotes the QFI with respect to $\hat{S}_z$. The corresponding phase sensitivity is given by

$$\Delta\phi^2 = \frac{\mathrm{Var}(\hat{P})}{\left(\partial_\phi \langle\hat{P}\rangle\right)^2} \approx \frac{\phi^2 F_Q/4}{(\phi F_Q/2)^2} = \frac{1}{F_Q}, \tag{S23}$$

where the variance is approximated by $\mathrm{Var}(\hat{P}) \approx \langle\hat{P}\rangle\left(1 - \langle\hat{P}\rangle\right) \approx \phi^2 F_Q/4$ for small ϕ. Therefore, the projection measurement of $\hat{P}$ saturates the QFI bound.

Instead, if we measure $\hat{S}_z$, the signal now is given by:

$$\langle\hat{S}_z\rangle \approx \frac{N}{2}(1 - \phi^2 \mathcal{C}_t) \quad \mathcal{C}_t = \langle\psi_0|\,\hat{S}_z^2(t)\,|\psi_0\rangle - \frac{2}{N}\langle\psi_0|\,\hat{S}_z(t)\hat{S}_z\hat{S}_z(t)\,|\psi_0\rangle\,, \tag{S24}$$

and for the variance

$$\mathrm{Var}(\hat{S}_z) \approx (\frac{N}{2})^2 \mathcal{V}_t \phi^2 \quad \mathcal{V}_t = 2\mathcal{C}_t^2 - \langle\psi_0|\,\hat{S}_z^2(t)\,|\psi_0\rangle + (\frac{2}{N})^2 \langle\psi_0|\,\hat{S}_z(t)\hat{S}_z^2\hat{S}_z(t)\,|\psi_0\rangle\,. \tag{S25}$$

In Fig. S3(b) and (c), we plot the rescaled quantities $\mathcal{C}_t/N^2$ and $\mathcal{V}_t/N^2$ as a function of time for different atom numbers. We observe that all curves collapse onto a single trajectory, indicating a universal behavior. Motivated by Eq. (S19), we propose the following simple ansatz $C\left(e^{-\frac{a}{(\chi_3 N^{3/2} t)^2}} - e^{-\frac{b}{(\chi_3 N^{3/2} t)^2}}\right)$, and fit the numerical results to obtain:

$$\mathcal{C}_t = N^2\left(e^{-\frac{7}{9(\chi_3 N^{3/2} t)^2}} - e^{-\frac{23}{9(\chi_3 N^{3/2} t)^2}}\right), \quad \mathcal{V}_t = 4N^2\left(e^{-\frac{10}{9(\chi_3 N^{3/2} t)^2}} - e^{-\frac{16}{9(\chi_3 N^{3/2} t)^2}}\right), \tag{S26}$$

as shown by the black curves. As illustrated in Fig. S3(b) and (c), these expressions provide accurate approximations for $\chi_3 N^{3/2} t \lesssim 1$.

Additionally, since $\mathcal{C}_t \propto N^2$, the condition $\phi \lesssim 1/N$ ensures that the protocol operates within the dynamical range where only lower-order terms contribute significantly. However, it is also necessary to choose a nonzero center phase to ensure a finite signal derivative and non-vanishing variance [S11, S12]. In the following, we denote such a center phase by ϕ_0. The metrological gain with respect to the standard quantum limit $\Delta\phi^2_{\mathrm{sql}} = 1/N$ can be calculated as:

$$G = \frac{(\partial_\phi \langle\hat{S}_z\rangle)^2}{N\,\mathrm{Var}(\hat{S}_z)} \approx \frac{4\mathcal{C}_t^2}{N\,\mathcal{V}_t} \approx N\frac{\left(e^{-\frac{7}{9(\chi_3 N^{3/2} t)^2}} - e^{-\frac{23}{9(\chi_3 N^{3/2} t)^2}}\right)^2}{e^{-\frac{10}{9(\chi_3 N^{3/2} t)^2}} - e^{-\frac{16}{9(\chi_3 N^{3/2} t)^2}}}, \tag{S27}$$

In Fig. S4(a), we plot the metrological gain versus atom numbers, showing performance close to the Heisenberg limit for a fixed center phase $\phi_0 = 10^{-4}$. In Fig. S4(b), we present the gain as a function of time for various atom numbers, also with $\phi_0 = 10^{-4}$. Figure S4(c) shows the optimal gain (maximized over time) as a function of ϕ_0, revealing a plateau that indicates the gain is nearly independent of the center phase for $\phi_0 \lesssim 1/N$, consistent with the analytical expression in Eq. (S27).

S3. DISSIPATIVE DYNAMICS

A. Exact numerical simulations

For the numerical simulations with single-particle spin flips, we use the exact permutationally invariant subspace method developed in Ref. [S13] to compute $\mathrm{Var}(\hat{S}_z)$ and the derivative $\partial\langle\hat{S}_z\rangle/\partial\phi$. To improve the accuracy of the derivative in the presence of dissipation, we rewrite the signal as

$$\langle\hat{S}_z\rangle = \mathrm{Tr}\left(\hat{S}_z e^{\mathcal{L}_{-\chi_3}t}\left(e^{i\phi_0\hat{S}_z}e^{\mathcal{L}_{\chi_3}t}(|\psi_0\rangle\langle\psi_0|)e^{i\phi_0\hat{S}_z}\right)\right), \tag{S28}$$

where $\mathcal{L}_\chi$ denotes the Lindblad generator with 3BI strength χ, including both single-particle spin flips and balanced superradiance. From this form, we observe that the derivative of the signal with respect to ϕ is given by the imaginary part of a two-point correlation function:

$$\frac{\partial}{\partial\phi}\langle\hat{S}_z\rangle = -i\left[\mathrm{Tr}\left(\hat{S}_z e^{\mathcal{L}_{-\chi_3}t}(\hat{S}_z\rho_t)\right) - \mathrm{Tr}\left(\hat{S}_z e^{\mathcal{L}_{-\chi_3}t}(\rho_t\hat{S}_z)\right)\right] = 2\mathrm{Im}\langle\hat{S}_z(2t)\hat{S}_z(t)\rangle, \tag{S29}$$

where we now define $\rho_t = e^{i\phi_0\hat{S}z}e^{\mathcal{L}\chi_3 t}(|\psi_0\rangle\langle\psi_0|)e^{i\phi_0\hat{S}_z}$. Throughout all numerical simulations, we evaluate the derivative using Eq. (S29).

For Fig. 4(c) in the main text, we apply the same framework to evaluate the metrological gain for both one-axis twisting (OAT) and two-axis twisting (TAT). The corresponding Hamiltonians and jump operators are summarized in the End Matter. For OAT, we simulate the time-reversal protocol proposed in Ref. [S14]. The system is initially prepared in a coherent spin state polarized along the x axis. It then evolves under the OAT Hamiltonian for a duration τ, followed by a small perturbative rotation about the $\hat{S}_y$ axis. Subsequently, the sign of the Hamiltonian is reversed and the system evolves for another interval τ. Finally, the observable $\langle\hat{S}_y\rangle$ is measured. The corresponding metrological gain is defined as $G_{\mathrm{OAT}} = (\partial_\phi\langle\hat{S}_y\rangle)^2/(N\mathrm{Var}(\hat{S}_y))$, with the derivative of the signal $\partial_\phi\hat{S}_y = 2\mathrm{Im}\langle\hat{S}_y(2\tau)\hat{S}_y(\tau)\rangle$ evaluated using two-point correlation functions. Similarly, for the TAT Hamiltonian realized in our system, $H_{\mathrm{TAT}} \propto \hat{S}_x^2 - \hat{S}_z^2$ [S6], the squeezing and anti-squeezing axes are given by $\hat{S}_s = (\hat{S}_x - \hat{S}_z)/\sqrt{2}$ and anti-squeezing along $\hat{S}_a = (\hat{S}_x + \hat{S}_z)/\sqrt{2}$. In this case, the perturbation is applied along the squeezing axis $\hat{S}_s$, and the signal is extracted from the measurement of $\langle\hat{S}_s\rangle$ [S15]. The corresponding metrological gain is therefore $G_{\mathrm{TAT}} = (\partial_\phi\langle\hat{S}_s\rangle)^2/(N\mathrm{Var}(\hat{S}_s))$ with the phase derivative given by $\partial_\phi\hat{S}_s = 2\mathrm{Im}\langle\hat{S}_s(2\tau)\hat{S}_a(\tau)\rangle$.

For Fig. 4(d), we present the metrological gain quantified by the quantum Fisher information (QFI), taking into account both the unitary dynamics and superradiant dissipation. For a general mixed state ρ governed by the master equation, the QFI associated with collective spin generators is evaluated using the spectral decomposition of ρ [S16]. In particular, the QFI matrix elements are given by

$$F_{ij}(\rho) = 2\sum_{k,k',\ q_k+q_{k'}>0}\frac{(q_k - q_{k'})^2}{q_k + q_{k'}}\langle k|\hat{J}_i|k'\rangle\langle k'|\hat{J}_j|k\rangle, \qquad i,j = x,y,z. \tag{S30}$$

where $\rho = \sum_k q_k|k\rangle\langle k|$ denotes the eigen-decomposition of the density matrix and J_i $(i = x,y,z)$ are collective spin operators. The maximal QFI, corresponding to the optimal measurement direction, is given by the largest eigenvalue of the 3×3 QFI matrix F_{ij}. However, instead of diagonalizing the full QFI matrix, we restrict the phase-encoding generators in our simulations to directions that are experimentally realistic for sensing applications in the main text. Specifically, for the 3BIs case we plot F_{zz}; for OAT with the initial state prepared along the z axis, we evaluate the QFI in the x-y plane; and for TAT with the initial state prepared along the y axis, we consider the QFI in the x-z plane. We find that the resulting values differ negligibly from those obtained by fully diagonalizing the QFI matrix, indicating that the restricted generator subspaces capture the nearly optimal metrological gain.

To ensure numerical stability in the presence of dissipation, we evaluate the QFI using a Lanczos-based partial diagonalization of the density matrix and discard eigenvalues below a fixed threshold. This procedure suppresses spurious contributions from near-zero eigenvalues in the denominator and yields a numerically robust estimate of the mixed-state QFI.

B. Atom loss

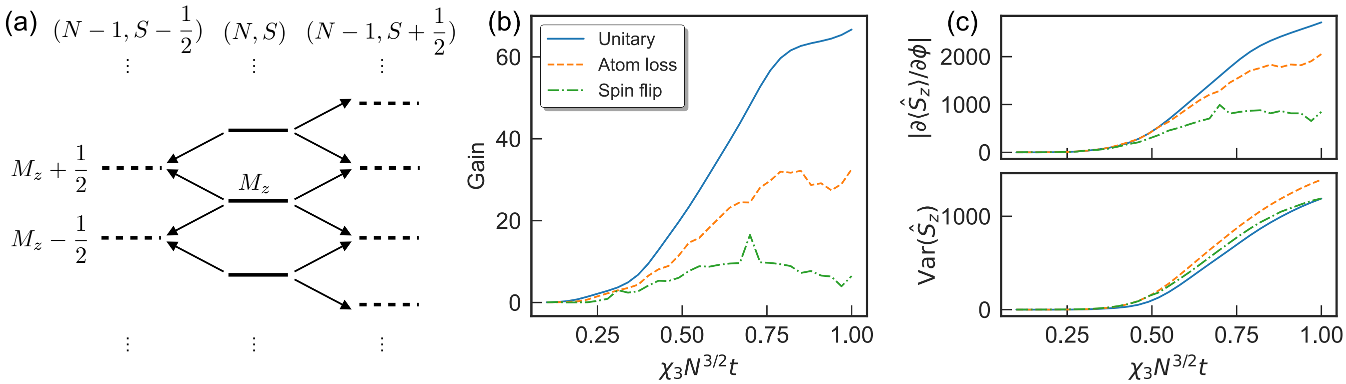

FIG. S5. Monte Carlo simulations of the dynamics of $N = 93$ atoms and $C = 10$ with atom loss processes. (a) Illustration of quantum jumps in the Dicke state manifolds due to atom loss. Each loss event couples the state (N, S, M_z) (solid levels) to $(N-1, S \pm \frac{1}{2}, M_z \pm \frac{1}{2})$ (dashed levels), with transition amplitudes given in Ref. [S17]. (b) Metrological gain vs evolution time for $\phi_0 = 1/N$. The orange dashed line shows the result with atom loss via the quantum jump method, while the blue solid and green dotted lines represent unitary evolution and with spin flip, respectively, both computed using ED. (c) Derivative and variance of the signal vs evolution time.

In this section, we incorporate atom loss, the dominant single-particle dissipation in matter-wave systems [S3]. This process corresponds to leakage from the pseudo-spin manifold into other momentum or internal states, and is modeled using jump operators $\hat{L}_i = \sqrt{\gamma_e}\hat{l}_i$ for the loss of atom i.

Between quantum jumps, each trajectory evolves under a non-Hermitian Hamiltonian, $\hat{H}_{\mathrm{NH}} = \hat{H}_3 - i\Gamma/2(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+) - i\gamma_e/2\sum_i \hat{l}_i^\dagger \hat{l}_i$, until a quantum jump occurs, determined by a random number. We only need to consider a fixed total atom number N and spin length S, thus reducing the memory requirement from $O(N^3)$ to $O(N)$ [S17]. When a quantum jump occurs, the state originally at (N, S, M_z) is coupled to $(N-1, S \pm \frac{1}{2}, M_z \pm \frac{1}{2})$ as shown in Fig. S5(a), with transition coefficient calculated in [S17]. Average over many trajectories provides an estimate of the expectation values of observables.

In Fig. S5(c), we show the time evolution of the derivative $\langle \hat{S}_z \rangle$ and the variance $\mathrm{Var}(\hat{S}_z)$ for $N = 93$, cooperativity $C = 10$, central phase shift $\phi_0 = 1/N$, and detuning $2\Delta_c/\kappa = \sqrt{NC/2}$. We find that single-particle atom loss leads to a larger amplitude in the derivative compared to spin-flip dissipation. Although atom loss slightly increases the variance, the resulting metrological gain remains higher, as shown in Fig. S5(b). A more systematic study of the impact of atom loss on metrological performance is left for future work.

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